

Technical Datasheet Sensor Modules with Modbus RTU Protocol with RS485 Interface

The Sensor modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital Sensor module is sturdy, low power usage and easy to use.

PT100/1000 Module: -



The Sensor modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The DIP switch for Slave ID and Baud rate are placed inside the enclosure.

Handles 100 Ω to 1k Ω (at 0°C) Platinum RTDs (PT100 & PT1000), Compatible with 2-, and 3-Wire Sensor Connections.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

Connectors	2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	70 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 – 24 VDC or 24 V AC/DC Typical – 12V DC @ 80mA
Operating Temperature	0 – 60 °C (32 ~ 140F)
Storage Temperature	-20 - 70 °C (-4 ~ 158F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



AI Inputs –

Channels	4
Input Signal	4 – PT100 / PT1000 – 10 V (jumper selectable)-Two wire/Three wire
Accuracy	Total Accuracy over All Operating Conditions: 0.5NC (0.05% of Full Scale) max.
Input Resolution	15-Bit ADC Resolution.
Isolation	Optically Isolated.

Additional Features: -

All inputs and communication ports isolated

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, □ 30KV contact, □ 30KV air EFT IEC 61000-4-4, 50A (5/50ms) 750V isolation.

CRC Error check.

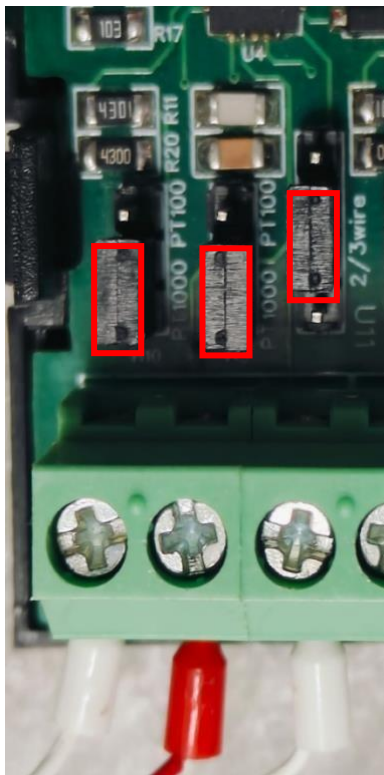
No configuration needed on the IO board

Configuration Settings: -

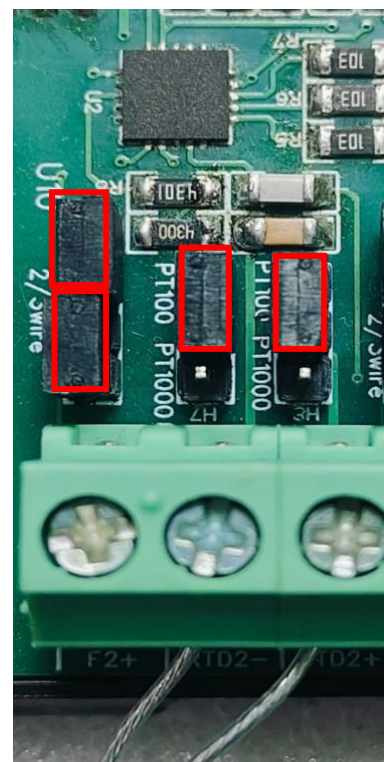
Communication Speed	9600 – 19200 Kbps (DIP SW Selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Configurable with DIP Switch
Function Code	0x03 Read Holding Registers
AI Register Address	0,1,2,3.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
0	RTD1	40001	0X03	RS485	16 Bit Unsigned int
1	RTD2	40002	0X03	RS485	16 Bit Unsigned int
2	RTD3	40003	0X03	RS485	16 Bit Unsigned int
3	RTD4	40004	0X03	RS485	16 Bit Unsigned int

2&3 wire Connections : -



3 wire connection

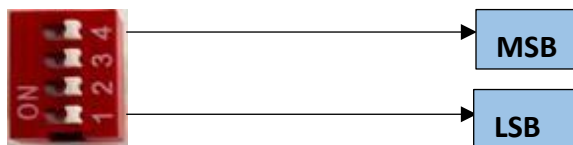


2 Wire connection

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

BAUD RATE DESCRIPTION

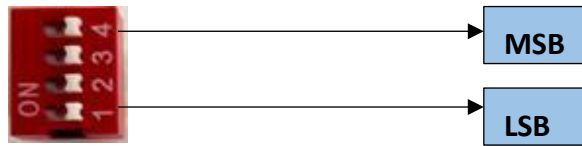


- For Baud rate Selection, DIP SW is used as per the diagram.
- Pulling up the switch will make Baud rate active.
- If no selection is made 9600 will be default Baud rate.
- When u change the Baud rate in the Module power 'ON' condition,

please press the reset button to get Change to affect.

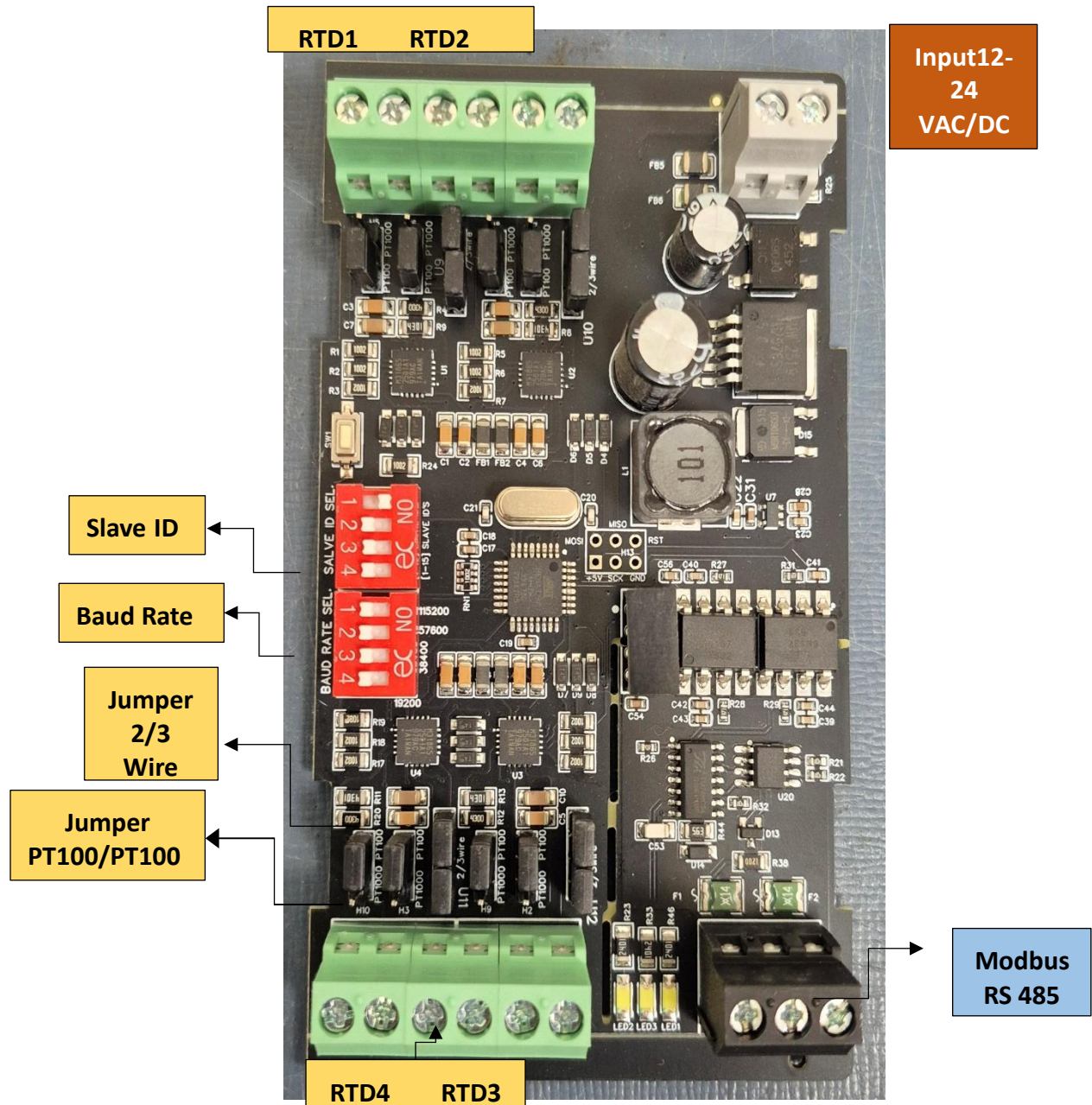
Baud Rate	DI P SWITCH			
	1	2	3	4
9600	OFF	OFF	OFF	OFF
19200	ON	OFF	OFF	OFF
38400	OFF	ON	OFF	OFF
57600	OFF	OFF	ON	OFF
115200	OFF	OFF	OFF	ON

SLAVE ID DESCRIPTION



- For Slave ID Selection SW is used to Set The SLAVE ID .
- For Slave ID DIP Switch **LSB is “1”** follow through **“4” is MSB**.
- Slave ID Confirmed through below Device ID table .
- IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So”1 0 0 0” will be selected as Slave ID 1.

Slave ID	DIP SWITCH				OUTPUT (Binary)	OUTPUT (Decimal)
	1	2	3	4		
0	OFF(0)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
1	ON(1)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
2	OFF(0)	ON(1)	OFF(0)	OFF(0)	0 0 1 0	2
3	ON(1)	ON(1)	OFF(0)	OFF(0)	0 0 1 1	3
4	OFF(0)	OFF(0)	ON(1)	OFF(0)	0 1 0 0	4
5	ON(1)	OFF(0)	ON(1)	OFF(0)	0 1 0 1	5
6	OFF(0)	ON(1)	ON(1)	OFF(0)	0 1 1 0	6
7	ON(1)	ON(1)	ON(1)	OFF(0)	0 1 1 1	7
8	OFF(0)	OFF(0)	OFF(0)	ON(1)	1 0 0 0	8
9	ON(1)	OFF(0)	OFF(0)	ON(1)	1 0 0 1	9
10	OFF(0)	ON(1)	OFF(0)	ON(1)	1 0 1 0	10
11	ON(1)	ON(1)	OFF(0)	ON(1)	1 0 1 1	11
12	OFF(0)	OFF(0)	ON(1)	ON(1)	1 1 0 0	12
13	ON(1)	OFF(0)	ON(1)	ON(1)	1 1 0 1	13
14	OFF(0)	ON(1)	ON(1)	ON(1)	1 1 1 0	14
15	ON(1)	ON(1)	ON(1)	ON(1)	1 1 1 1	15



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